Letter: Bureaucratic Influence

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Abstract

Some bureaucrats are especially able to shape agency outputs. Yet, existing empirical measures of bureaucrats' influence miss the richness of who holds sway in public organizations. We propose a new dynamic measure of bureaucratic influence: email response time. We contend that bureaucrats perceived by colleagues as powerful or possessing important information will be prioritized and responded to sooner. We validate this measure using data on all internal emails sent by employees in Texas' public health agency over four one-week periods in 2020 and 2021, demonstrating that within-employee response rates are faster when emails are sent by senior and higher-earning employees. We then display the measure's usefulness by testing whether senior managers' or scientists' emails were responded to more promptly after the onset of the COVID-19 pandemic. We conclude by discussing the applicability of this measure to other contexts, such as the marginalization of opposing civil servants.

Keywords: Bureaucracy, Influence, Agency Politicization

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The opportunity to influence public policy is an attractive feature of public service. Bureaucrats, unlike their peers in the private sector, are directly involved in governance. They draft health regulations, conduct environmental inspections, procure defense equipment, and approve lifesaving medications. Yet, not all bureaucrats have the same degree of influence over agencies' outputs. Whether due to their position in the organizational hierarchy, unique topical expertise, ideological alignment with the executive, or valuable relationships with industry or political actors, some public employees have a greater influence than others over what agencies ultimately do.

Organizational charts suggest that top agency heads and their deputies possess the most power over agencies' outputs, but laws, regulations, and other agency actions are not simply dictated by those at the top of an agency's hierarchy. Long-serving civil servants and mid-level political appointees can serve as "spigots" or "choke points" within their agencies (Heclo 1977; Doherty, Lewis, and Limbocker 2019a; Lewis 2011). Likewise, rule drafters can impact the timing and content of regulations (Potter 2017; Bellodi et al. 2026), and considerable research has shown that agency outputs are in part shaped by the frontline, street-level employees charged with implementing policy (e.g., Lipsky 2010).

Relatively little research has measured who is influential in an agency and how that influence waxes or wanes over time. Surveys of federal employees have measured which groups of employees are perceived as being especially influential (Bertelli and Lewis 2013; Richardson 2019) and studies have examined how agencies themselves gain independent policymaking power (Carpenter 2001), but we are unaware of any studies that have attempted to systematically measure individual bureaucrats' influence or how it fluctuates over time. We innovate by proposing a new dynamic measure of bureaucratic influence: email response times. Building on principles from social psychology and organizational management, we argue that bureaucrats, like private sector employees, are incentivized to cultivate a positive image with their supervisors and other powerful colleagues. They are also pressured, due to time and effort constraints, to prioritize communications from peers with consequential information. Both mechanisms, we contend, point toward bureaucrats prioritizing the messages of colleagues viewed as occupying important positions within an agency.

We validate our measure using an original dataset of over 385,000 email messages sent by employees of the Texas Department of State Health Services (DSHS)—the state's public health agency—over four week-long periods in 2020 and 2021. These data include the metadata—the time the email was sent, the email address of the sender, and the email address(es) of the receiver(s)—of all internal emails sent by DSHS employees over the four periods, providing a thorough glimpse into communication networks within the agency. We match the email metadata records to employees' personnel records by matching email addresses to employee names. The personnel data, also received via public records requests, contain the name, race, sex, salary, start date, and job title of DSHS employees.

Matching emails to their subsequent responses, we first validate our measure by regressing, via a Cox survival model, email response time against established, albeit imperfect, measures of influence: seniority and salary. We show that emails sent by higher-earners and more highly-ranked bureaucrats are responded to at a faster rate, especially among those messages left unanswered after at least one hour. This suggests, we contend, that repliers prioritize the messages of colleagues viewed as more influential. We then move to displaying the applicability of our measure by testing whether employee influence changed after the onset of the COVID-19 pandemic. Using a similar survival design we find little consistent evidence that either scientists or senior managers in the DSHS enjoyed faster responses to their emails after the onset of the COVID-19 pandemic.

We close by discussing the applicability of our measure to other questions in executive politics. Recent research on bureaucratic turnover following the election of a new president has frequently found little evidence of departures among misaligned civil servants with different policy preferences (Spenkuch, Teso, and Xu 2023; Bolton, Figueiredo, and Lewis 2021; Doherty, Lewis, and Limbocker 2019a). One possible source of this largely null finding is that departing the workforce is a rare event. More fine grained measures of influence that can pick up on losses of influence over time offer a window for examining whether marginalization is still occurring, even if it does not lead to observable departures from the public sector.

1 Background

Hierarchy is a defining feature of bureaucracy. Agencies' power to write and implement regulations, fight wars, and enforce laws flows from political leaders and statutes. Within an agency, organizational charts structure who reports to whom and the respective authority and responsibility of each position. As Weber (1946, p. 197) puts it, bureaucracy is in part characterized by "office hierarchy and. . . levels of graded authority" consisting of "a firmly ordered system of super- and subordination in which there is a supervision of the lower offices by the higher ones." In another seminal account, Downs (1967, p. 49) lists a "hierarchical structure of formal authority" and "hierarchical formal communication networks" as two characteristics that all agencies have in common. Roles in an agency are ostensibly well-defined, with everyone aware of who they report to and who reports to them. Likewise, influence over agency outputs is descending with rank. The highest leadership positions in an agency possess the greatest influence, while the entry-level clerks and frontline employees possess less power to decide policy. Lower-level positions can and do make important decisions but those choices can be reviewed and altered by more powerful supervisors.

Yet, hierarchy does not always cleanly map to influence over agency outputs. High-level officials in an agency often cannot simply dictate policy changes. They have to delegate some level of power to lower-level bureaucrats with the information, expertise, or time to complete tasks (Nou 2017). For instance, Luis Aguilar, the former Commissioner of the Securities and Exchange Commission (SEC), wrote in an outgoing message to future Commissioners that "from time to time, you might read in a newspaper about a "Commission action," and you will have no idea what it is about. . . In all likelihood, the staff had taken action pursuant to the more than 376 separate rules where the Commission previously granted delegated authority to the SEC staff. (Aguilar 2015)" These delegations of power within agencies can generate principal-agent problems within public organizations. Lower-level employees are at times able to utilize informational advantages to shift policy outcomes toward their preferred outcomes. While political appointees can set the direction of policy, they are limited in their ability to fully implement their priorities. Other bureaucrats within the agency, not only top leadership, enjoy influence

over policy.

Politicization—the degree to which executives, such as presidents, governors, and mayors, have gained control over agency policymaking via appointees and other tools—is a key factor shaping the relative balance of influence between high-level political appointees and career staff (Lewis 2008, 2011; Limbocker, Richardson, and Selin 2022). Agencies that are more politicized have greater policymaking influence centralized in political appointees, relative to more independent agencies where career employees enjoy greater power. Politicization inherently involves changing levels of policy influence and, as a result, the best measures of who enjoys influence in agencies occur in studies of agency politicization. In their study of bureaucratic turnover, Bertelli and Lewis (2013) show that as senior civil servants perceive that they have less influence relative to political appointees, they are more likely to want to leave their agency. They measure employees' perceived level of influence by asking senior federal civil servants: 'In general, how much influence do you think the following groups have over policy decisions in [employee's agency]?" The employee groups included, among others, "senior civil servants" and "political appointees" and respondents are able to answer with "a great deal," "a good bit," "some," "little," "none," and "don't know." Bertelli and Lewis (2013) use this measure of relative influence to construct their measure of interest: the difference in the perceived influence of political appointees and senior civil servants. Richardson (2019) uses the same measure in his related study of the effects of politicization on turnover and effort, demonstrating that higher relative influence of political appointees is correlated with less expertise acquisition by senior civil servants.

Survey-based measures of influence offer insights into how bureaucrats' perceptions of who wields power in an agency affect their own behavior. But they also have drawbacks. First, existing survey-based measures of policy influence are aggregated. Both Bertelli and Lewis (2013) and Richardson (2019) ask bureaucrats to weigh the influence of all senior civil servants against all political appointees. As a result, it is not possible to parse how, within a hierarchical level, which civil servants or political appointees are viewed as more influential. This is important because politicization likely does not equally affect all employees. For an executive interested in gaining control of an agency, they are incentivized to reduce the influence of opposing civil servants but other indifferent or aligned civil servants within the same agency may actually receive ad-

ditional grants of discretion. Unlike their peers in the agency, aligned civil servants might find that politicization empowers them with additional policy influence.

Second, influence is dynamic, which is difficult to fully capture via expensive survey instruments. Politicization is inherently a longitudinal process. New executives gain office and, observing the alignment of various agencies and taking stock of their own preferences, increase their control over certain agencies. Static measures of influence struggle to fully pick up on these trends. While surveys can be run multiple times, the costs and logistics of surveys – especially those focused on unique populations like civil servants – make this approach difficult. Third and finally, survey-based measures of influence are subject to the usual challenges associated with survey instruments, such as social desirability bias. While Richardson (2019) is careful to show that his results are not a product of such bias, the problems are still an ever-present possibility whenever survey instruments are utilized.

Extant measures of who is influential in an agency provide a helpful window into intra-agency policymaking dynamics. But they are also imperfect. Influence is dynamic and fine-grained, varying not only across civil servants and political appointees but also within position type. Some civil servants , whether due to their alignment with the administration, unique expertise, or some other factor, are likely more influential than some of their civil servant peers. Existing measures cannot pick up on this variation. Likewise, existing measures struggle to pick up on the waxing and waning of influence over time. Major changes to an agency's context, such as the election of a president with different policy priorities or an external event that sharply changes agency tasks, offers opportunities for some employees to become more or less influential. Fully picking up on these measures requires a measure of bureaucratic influence that is both measured at the individual level and is able to be measured over time.

2 Influence as Prioritization

Many bureaucrats enter public service because they want to affect policy (Gailmard and Patty 2007, 2012; Perry 1996; Perry, Hondeghem, and Wise 2010). Working within government agencies offers an opportunity to develop regulations, provide public services, and implement

government programs. Yet, bureaucrats are not immune from other professional goals (Downs 1967). Like their private sector counterparts, bureaucrats prefer higher levels of financial compensation, as well as the benefits and job security offered by many public sector positions (Lee and Whitford 2008; Triest 2024).

Regardless of a bureaucrat's particular motivations, retaining or advancing in one's career is a key means for achieving them. Promotions offer greater opportunity to influence public policy, as well as improved financial compensation. Even if an employee is unmotivated by greater financial rewards and content with the security and benefits of their current civil service position, they will likely at least be motivated to keep their job. While civil service systems make it more challenging for supervisors to remove poorly performing employees, dismissals on grounds of performance still occur. In addition, an increasing number of public employees in the U.S. states and federal government lack due process rights and other protections, increasing the likelihood that poor performance will be met with professional consequences (McGrath 2013; Schedule F 2020; Schedule Policy/Career 2025; Goehring 2025).

Performing well on the job is an important driver of career advancement in public bureaucracies (Oh and Lewis 2013), but doing a good job at work requires the allocation of a limited amount of time and effort. Bureaucrats, like anyone else, cannot work all day and night. They are limited in the amount of time they can spend on an assignment before needing a coffee break. Oftentimes, they likely cannot complete every task and send every email in a given day. As a result of these limitations, they need to be strategic in how they spend their time and energy. This is especially the case in professions where resources are limited and stress levels are high, such as teaching and correctional work, where employees have to develop practices for "coping" on the job (Tummers et al. 2015). Regardless of the specific circumstances, though, bureaucrats need to prioritize tasks, assignments, and colleagues in order to most effectively perform their job. They need to decide how to allocate their time to perform well and advance, or at least retain, their jobs.

In order to perform their jobs well, we contend that bureaucrats are going to prioritize tasks and communications from two groups of colleagues: 1) supervisors who one is accountable to;

and 2) colleagues with information that is necessary for making important decisions. First, bureaucrats have considerable incentives to prioritize their supervisors and other colleagues who can decide future career trajectories. Research from social psychology and management has shown that employees work to manage how they are viewed by their colleagues, especially powerful ones. This can take different forms, including expending effort that serves the best interests of the organization as well as cultivating an image of dedication in the eyes of supervisors who one is accountable to (Bolino 1999; Tetlock 2002; Bolino et al. 2006; Wayne and Ferris 1990). These are not mutually exclusive strategies, and research has found that, regardless of its authenticity, cultivating a positive image in the eyes of supervisors correlates with improved performance reviews (Higgins, Judge, and Ferris 2003; Peck and Levashina 2017).

Prompt responses to communications and requests from senior colleagues displays an employee's attentiveness and responsiveness, likely improving one's image in the eyes of influential supervisors. Fast responses are associated with other positive outcomes in communication contexts outside of email. Templeton et al. (2022), for instance, finds that fast responses in conversation leads to greater feelings of social connection. Inversely, Ziano and Wang (2021) shows across a number of different social interaction contexts that responding slowly to someone else leads to feelings of insincerity.

We also expect that bureaucrats will prioritize peers who possess important information. Bureaus are large, impersonal organizations where not every employee knows everything about the tasks of an agency (Downs 1967). Performing well on the job requires, in part, an ability to recognize who in an agency possesses important information, acquire that information, and utilize it on the job. For example, as we discuss more later, COVID-19 required changes in many bureaucrats' jobs. Individuals were pulled onto new projects and tasks, setting up infection-tracking programs, social distancing guidelines, and other public health measures. Supervisors in many agencies needed information in order to make the correct decisions on these consequential policy issues. As a result, they turned not only to external sources but also internal employees — scientists, data scientists, and other public health specialists. In order for the supervisors to perform their jobs well, in other words, they needed to work closely with peers who possessed necessary expertise.

Both of these expectations have observable implications: employees perceived as more important, whether due to their status or expertise, will enjoy greater responsiveness. We propose measuring this responsiveness via the elapsed time between an email being sent and replied to. Importantly, this is a measure of relative, not absolute, influence. It might be the case, for instance, that an especially influential employee supervises a group of employees who respond slowly to all emails. In contrast, a less-influential employee might consistently interact with fast repliers. Comparing the response time of these two employees to one another would generate an incorrect assessment of who bears more influence in the agency. Instead, we expect influence to matter within a given replier—for a given employee, we expect them to respond quicker to more influential colleagues.

Measuring responsiveness via email response times has numerous advantages. For one, email is a ubiquitous form of workplace communication. While other options for online professional communication exist—such as Slack, Microsoft Teams, and other instant messaging platforms—email is still widely used, with one recent estimate suggesting that the average employee receives 117 emails per day (Microsoft 2025). The omnipresence of email means that, as a measurement tool, email response times are not isolated to a specific subset of employees. Both higher-level officials and lower-level staff utilize email, meaning that email response times can be measured across occupations and hierarchical positions. In addition to its comprehensiveness, email is a useful tool for measuring responsiveness because it is both fine-grained and longitudinal. Email response times are measured in minutes and seconds, not weeks or months, meaning that comparisons can be made at very precise levels. Additionally, unlike surveys, email data can be analyzed at more than one point in time (or as a continuous flow), offering a window into how response rates evolve after key events like elections or pandemics.

Data accessibility is another advantage of email-based influence measures. As we discuss further below, comprehensive email metadata snapshots are available from at least some U.S. state agencies via public records requests.[1] These records lack any information on the subject or body

1. Anecdotally, we have found considerable variation across and within states in agencies' willingness to provide these metadata records. It appears that some state agencies' IT systems make accessing these data very easy, while other systems make it impossible. As far as we know similar metadata are not available from federal agencies or presidential archives.

of the email. This is a necessary drawback. Requesting any communication text requires public records officials to perform extensive redactions, leading to exorbitant costs, excessive wait times, and considerably restricted data availability.

While we are the first (that we know of) to study bureaucrats' email response times, other scholars outside of political science and public administration have examined email response patterns. In a large study of over 2 million email accounts, Kooti et al. (2015) shows that younger people respond more quickly to messages and overall response times decrease when messages are sent during working hours. Creamer et al. (2022) use average email response time as an input in their model of corporate hierarchy, showing that it correlates with email importance, although the relationship disappears after controlling for volume. Perhaps most relatedly, Gloor et al. (2017) finds that private-sector managers a few months away from leaving their jobs send more email nudges to induce replies, perhaps indicating decreased responsiveness from colleagues.

3 Validation

As a first step in validating our proposed measure of bureaucratic influence, we collected email metadata from all employees in Texas' public health agency, the DSHS. The records contain the email address of the sender, the email addresses of the receiver(s), the email addresses of any employees carbon-copied (cced) or blind carbon copied (bcced) on the message, and the date and time the email was sent (down to the second). We collected the metadata of all email communications sent by DSHS employees to other DSHS employees over four 7-day periods: January 3-9, 2020; March 13-19, 2020; September 11-17, 2020; and March 12-18, 2021. All four periods span from a Friday through the following Thursday. We chose these periods to get observations from both before and after the COVID-19 pandemic, which was declared a pandemic by the World Health Organization on March 11, 2020. Despite the relatively few columns in the dataset, the records still require cleaning. Email archival systems pull the same email more than once from the sender's and receivers' inboxes, resulting in considerable duplicates. While this is easy to diagnose when the sender, receiver, and sent time are identical, in some cases the sent time can be off by a few seconds across inboxes. Therefore, we deem any records from a given

sender to a given receiver within the same 5 seconds to be the same email.

After cleaning, the dataset includes 385,710 email messages across the four week-long periods. We match the email addresses in the metadata to employee personnel records received via public records requests to the state's human resources agency, the Texas Workforce Commission. These records contain employees' names, unique employee identification codes, race, sex, job titles, agency start dates, and salaries. We match the email addresses to the personnel records using crosswalks provided by the DSHS and additional name matching. In total, we were able to match 85% of all email messages to both the sender's and receiver's personnel files.

We calculate email response time by measuring, for each sender-to-receiver communication, the elapsed time in minutes before the receiver responds to the sender.[2] We drop receivers who are cced or bcced on a message as well as messages sent to more than 5 people to remove group communications that do not require a reply. Approximately 70% of the emails were matched to a subsequent reply. In the SI in Figure A.1 we show that the reply rate decreases with time within a given weekly data snapshot, indicating, as expected, that messages sent later in the respective week-long periods are less likely to receive a reply.

Table 1 provides basic information about the merged dataset. Within each week-long snapshot there are over 2,500 unique email recipients and senders, respectively. Email traffic—both sent and received—increases slightly after the onset of the COVID-19 pandemic in March 2020. Across all weeks, email traffic is right-skewed. While the median employee received four emails per day in the period March 12-19, 2021, the mean employee received eight, indicating that a small number of employees received a high number of messages from their colleagues. Right-skew is even more prevalent in email response times, which are shown in hours in the final two columns of Table 1. Across snapshots, the mean email response time, among emails with a reply, was over eleven hours. But this is driven by a small number of emails. Half of all emails were responded

2. A key shortcoming of this approach, that we plan to fix in future drafts, is that we are unable to track responses to specific messages. Instead, we use as the "reply" the first instance that the receiver of the original email sends an email back to the original sender. This means that, when a sender sends multiple emails before a replier responds, the response is counted as the reply to all of the sent emails. Since greater email traffic likely also correlates with influence, this is likely biasing our validation estimates below. We are collecting additional data on email threads that will let us track replies to specific messages.

Table 1 – Descriptive Information. Displays counts and summary statistics about employees, email traffic, and email response times.

	Employees		Emails					
	Receivers	Senders	Sent (day)		Received (day)		Resp. Time (hrs)	
Snapshot	N	N	Mean	Med.	Mean	Med.	Mean	Med.
Jan. 3-10, 2020	2,633	2,540	6	4	6	3	12.9	1.2
Mar. 13-20, 2020	2,700	2,606	8	4	7	4	11.3	0.8
Sep. 11-18, 2020	2,771	2,790	8	5	8	4	13.2	1.0
Mar. 12-19, 2021	2,831	2,841	8	5	8	4	12.7	1.0

to within approximately one hour. Notably, email response times ticked downward in March 2020 immediately after the WHO declared COVID-19 a pandemic before rebounding to pre-COVID-19 levels. This could be due to public health employees responding more urgently to messages due to the seriousness of the pandemic.

If email response time decreases with sender influence, then we should expect that senders higher in the organizational hierarchy will, all else equal, be responded to more quickly than peers in frontline positions. Even if hierarchy does not perfectly capture bureaucratic influence, it is certainly positively correlated. We test whether email response time decreases with two markers of organizational rank: salary and within-job class rank. Regarding the latter, nearly all positions in the DSHS have an associated rank (e.g., Epidemiologist I, Epidemiologist II, and Epidemiologist III). We treat the numeric components of these job titles as measures of an employee's ranking within the given position, offering another indicator of an employee's status within the DSHS hierarchy.

We test whether salary and rank correlate with faster email response time using a Cox regression model. Survival modeling is especially appropriate for analyzing time-to-email-reply. Since the email data are from snapshots in time, they are censored on both the right and the left. We do not observe the beginning of all email threads, nor do we observe all replies. In the Cox survival analysis, the outcome is the elapsed time in minutes between an email being sent and the email being replied to—with emails lacking a reply in our time frame being coded as censored. We include direct controls for whether a sender is white or male. We also stratify on a number

of variables—the hour of day an email was sent, the week-long snapshot, whether the email was sent on a Saturday or Sunday, and the sender's years of experience—estimating a separate baseline hazard of reply for each combination of values.[3] Finally, we include recipient-level strata to ensure that we are accounting for baseline differences in individual repliers' time to reply.

A core assumption of a Cox model is that hazards are proportional. That is, the ratio of the "risk" of an email receiving a reply when, for instance, it was sent by a high versus low earner is the same whether the email was left unanswered for ten minutes or ten hours. Tests of this assumption with our data suggest that proportional hazards do not hold for our two coarse markers of influence, salary and rank. Therefore, we test the relationship between email response speed and, respectively, salary and rank among emails that have gone unanswered for 0-60 minutes, 1-8 hours, and more than 8 hours.

Table 2 shows the coefficient estimates as hazard ratios with 95% confidence intervals in brackets. The first two columns use the full email dataset to regress email response time against, respectively, senders' monthly salary and rank. The first column aligns with expectations: Across time bins, emails sent from employees with higher salaries enjoy faster response times. Among emails unanswered in the first 60 minutes, a one standard deviation increase in sender salary correlates with a 5% higher risk of replying. The magnitude of the correlation is higher among emails still left unanswered after 60 minutes. For emails still unanswered one to eight hours after being sent, a one standard deviation increase in salary correlates with a 14% higher hazard of replying. The relationship between email response time and rank in column two is less clear. An increase in a sender's rank correlates with a faster response time among emails left unanswered after one eight-hour work day, but there is no statistically significant relationship across the two other time bins. One possibility for the differences across the salary and rank variables is that salary is a more universal measure of importance. A Program Manager III might not be viewed as being particularly important by a Manager II, for instance, even though the former has a higher rank.

3. Years of experience is calculated as the difference in years between the date of the data and the employee's hire date. For unknown reasons, the earliest hire date listed in the personnel records is September 2004, effectively censoring the years of experience variable. We bin years of experience into three categories: 0-5, 6-10, and 11+ years.

Table 2: Email Response Times Largely Positively Correlated with Extant Influence Measures. Shows the results of three separate Cox regression models that test for an association between a sender's position within the DSHS hierarchy and the speed with which repliers' respond to the message. Coefficients are displayed as hazard ratios with 95 percent confidence intervals.

	All Emails		Within-Position
	(1)	(2)	(3)
Sender's Monthly Salary (1 SD) × 0-60min	1.05		
	[1.03, 1.07]		
Sender's Monthly Salary (1 SD) × 1-8hr	1.14		
	[1.10, 1.18]		
Sender's Monthly Salary (1 SD) × 8hr+	1.14		
	[1.11, 1.18]		
Sender's Rank × 0-60min		1.01	1.04
		[0.99, 1.02]	[1.01, 1.07]
Sender's Rank × 1-8hr		1.00	1.14
		[0.98, 1.03]	[1.09, 1.20]
Sender's Rank × 8hr+		1.03	1.08
		[1.01, 1.05]	[1.01, 1.15]
Sender is Male	0.97	0.98	0.94
	[0.93, 1.00]	[0.95, 1.02]	[0.89, 1.00]
Sender is White	1.00	1.02	0.96
	[0.97, 1.04]	[0.99, 1.05]	[0.90, 1.01]
N	221 512	220 000	59 509
Hour-of-day Strata	✓	✓	✓
Snapshot Strata	✓	✓	✓
Weekend Strata	✓	✓	✓
Sender's Years Exp. Strata	✓	✓	✓
Receiver Strata	✓	✓	✓

Therefore, the third column provides a cleaner test of the relationship between rank and time to reply. This model restricts the data to only include communications between employees in the same job class (i.e, a Program Manager II emailing a Program Manager I or an Epidemiologist I emailing an Epidemiologist III). The advantage of this approach, in contrast to the model in column two, is that the hierarchical ranking of employees is especially clear. Using this approach, there is a large and statistically significant correlation between sender rank and response time. Among emails left unanswered for up to one hour, a one unit increase in the sender's rank correlates with a 4% higher risk of replying. That correlation increases in magnitude to 14% for emails left unanswered for 1-8 hours.

4 Application: Changing Influence After COVID-19?

We view the dynamic nature of our measure of bureaucratic influence as a key benefit. Email traffic is constant, meaning it can be captured and analyzed before and after events and shocks that potentially affect which bureaucrats are viewed as influential within an agency. In the Discussion we discuss the applicability of our measure to analyzing the marginalization of opposing civil servants after the election of a new executive. We focus on a different case in this section: the COVID-19 pandemic. Public health agencies, such as the DSHS, were on the front lines of the pandemic response in 2020, 2021, and beyond. Bureaucrats had to set up centers for mass testing, create systems to track infections and share the information with the public, provide expert epidemiological guidance to political leaders, and procure essential supplies like masks and ventilators. These were dramatic changes in the responsibilities of many public health agencies and their employees. Many bureaucrats in public health roles had to change their roles due to the pandemic, resulting in the diversion of resources away from what were, until the pandemic, core public health responsibilities, such as education and environmental health (Kintziger et al. 2021; National Association of County and City Health Officials 2022)

These changes to public health agencies' and bureaucrats' tasks likely had impacts on who was viewed as influential in an agency. While superordinate and subordinate relationships remained fixed, the importance of certain types of skills and expertise changed. Scientists, for instance, had insights into the severity and spread of the disease. While they certainly had in-

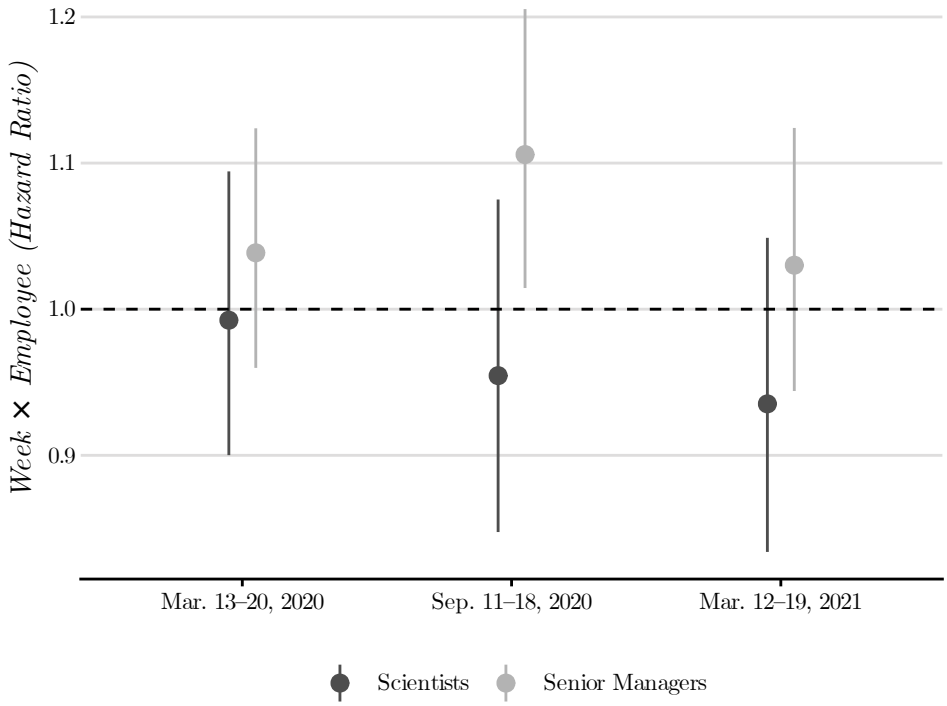
fluential roles before the pandemic, the weight with which their peers and supervisors put on that information likely shifted after the onset of the pandemic. This points to scientists becoming more influential due to the pandemic, which we contend will be able to measured as faster responses to scientists' emails.

On the other hand, the response to COVID-19 was hardly apolitical. Public health agencies faced pressure from political leaders and other actors to minimize disruptions to the economy and keep schools and places of worship open. This points to different expectations. Scientists, rather than becoming more influential, were actually marginalized as their work become more salient and polarizing. Another possibility along these lines is that senior agency leaders actually became more influential to COVID-19. Bureaucrats, rather than needing more scientific information to properly do their jobs required more strategic direction from the highest levels of the agency hierarchy.

We provide an initial descriptive test of these claims by examining whether emails sent by, respectively, scientists and senior managers in the DSHS were responded to more quickly after the onset of the COVID-19 pandemic.[4] We follow a similar Cox survival analysis design as above but, for simplicity, avoid estimating our model using different timing bins. The main coefficients of interest are the interactions between whether an employee is, respectively, a scientist or senior manager and an indicator for the week-long snapshot. The omitted category for the week-long snapshot variable is January 3-10, 2020—the period before COVID-19 was a prominent issue in the U.S. We include similar demographic, time-of-day, and time-of-week controls as in Table 2. Finally, we exclude emails sent to other employees within the same respective job class (e.g., emails sent from scientists to other scientists).

4. We define scientists as all employees in the following positions: microbiologist, molecular biologist, chemist, or epidemiologist. Managers include all employees with the job titles: mgr, director, commissioner dshs, or deputy director. In our dataset, 310 scientists and 287 senior managers sent at least one email.

Figure 1 – Change in Email Response Rates for Scientists and Senior Managers, relative to January 2020 and other employees. Shows coefficient estimates from two separate models that test the interaction between employee (scientist or senior manager, respectively) and the week-long snapshot. The interaction terms are exponentiated and are displayed as hazard ratios with 95% confidence intervals. Full tabular results are shown in Table B.1.



We plot the main coefficients of interest in Figure 1 and include the full tabular results in the SI in Table B.1. Each coefficient shows, along with the corresponding 95% confidence interval, the difference in the hazard of replying to a scientist (dark gray) or senior agency manager (light gray) in the given period relative to the pre-COVID January period. Overall, we find little consistent evidence of a change in email reply speed, relative to the pre-COVID January period. Across all three periods, there is no evidence of increased responsiveness to scientists in the DSHS. We also not do consistently find that senior managers' emails were responded to more quickly following the onset of the pandemic. The one exception is in September 2020, when emails from senior leadership enjoyed an approximately 10% greater hazard of replying. One possible reading of this result is that in the fall of 2020 the state was in the process of reopening. On September 17, 2020—during our study period—Governor Abbott issued an executive order

allowing for businesses, gyms, museums, libraries, and other spaces to reopen at 75% capacity (Abbott 2020). He specifically mentions that the order was issued in consultation with, among other actors, the then-Commissioner of the DSHS, John Hellerstedt. One possibility is that the process of drafting that order could have led to greater centralized control of the agency from top managers, like the Commissioner.

5 Discussion

Who is influential in a bureaucracy? We have posited one answer: bureaucrats who are prioritized by their peers. In this paper we set out to justify this approach theoretically and empirically, offering an initial validation of our measure using email metadata from Texas' public health agency. While more work needs to be done to justify our measure, we provide suggestive evidence that email response time correlates with coarse measures of influence, like employee salary and job class rank. Despite little consistent evidence of changes in senior managers' or scientists' influence following the onset of the COVID-19 pandemic, we are interested in pursuing other applications of our measure in contexts where changes in influence seem likely.

One potential application of our measure of bureaucratic influence is in the context of the marginalization of opposing civil servants. A number of studies have examined the effects of presidential transitions on civil servant turnover, testing whether civil servants with different policy preferences than the incoming administration are more likely to depart the workforce (Spenkuch, Teso, and Xu 2023; Bolton, Figueiredo, and Lewis 2021; Doherty, Lewis, and Limbocker 2019a, 2019b). In many of these studies, marginalization and/or an unwillingness to work for an opposing administration are the theoretical mechanisms explaining why misaligned civil servants might be more likely to depart following an election. A new executive—usually president—takes office and, given the difficulties of outright removing bureaucrats protected by civil service due process rights, opt to push opposing employees out of the workforce by assigning them to do jobs they do not want to do or work in places that they do not want to work. By and large, however, these studies have found little evidence of increased turnover of misaligned career bureaucrats following the inauguration of a new president from the opposite political party. While misaligned political appointees cycle in and out of government with the

election of a new administration, no similar trends exist among civil servants.

While there is little evidence of a political cycle in civil service turnover, that does not mean that there is no political cycle in civil service marginalization. It could be that the outcome variable (departing the workforce) is too coarse of a measure to pick up on marginalization by an incoming administration. Opting to leave one's job is a significant decision that many employees will likely be unwilling to make, even if they are being pressured by top agency leaders to depart. Furthermore, some of the most misaligned bureaucrats might be unwilling to leave in order to retain their influence over policy for when a more agreeable executive is elected (Cameron and Figueiredo 2020). In short, focusing on turnover misses that marginalization might still be occurring.

Email response times provide a much more fine-grained measure of marginalization (i.e., declining influence over policy) than leaving the public sector workforce. Even if marginalized employees do not depart the workforce following the election of a new executive, it might be the case that they do lose influence over policy. The political appointees in the new administration might not include them in important policy discussions or assign them to do tangential work that is less central to the agency's mission. No longer performing core policymaking duties, these opposed civil servants will be more isolated from top agency decisionmakers—and be less of a priority as a result. Email response times, we contend, can help pick up on these microlevel trends in changing influence over time within public agencies.

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Supplementary Information

Letter: Bureaucratic Influence

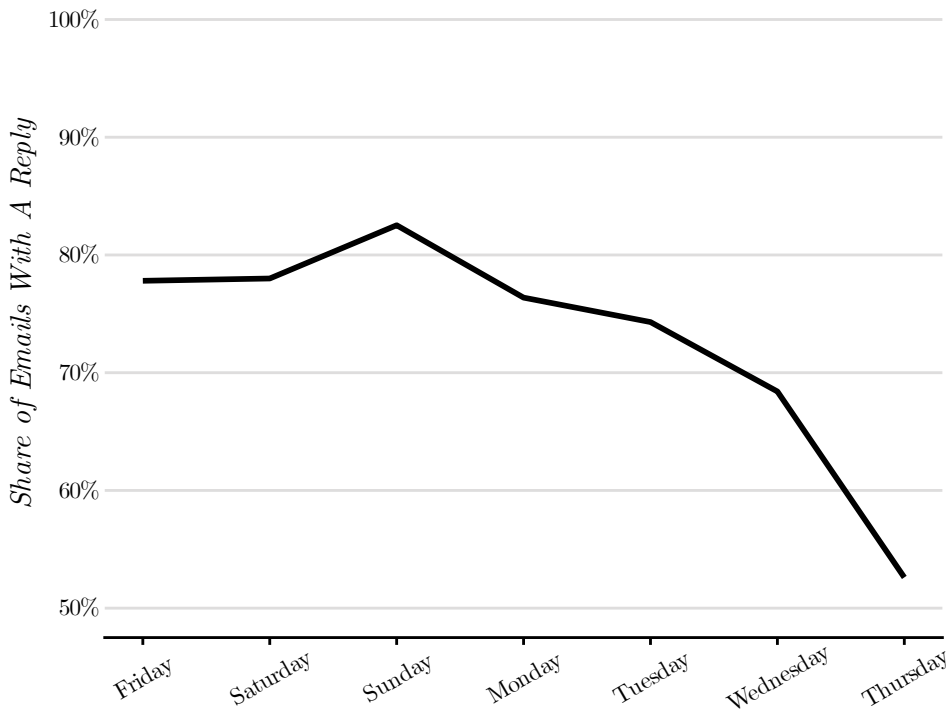
Benjamin Goehring and Adam Rauh

A Additional Descriptive Information **SI-1**

B Additional Results **SI-1**

A Additional Descriptive Information

Figure A.1 – Email Reply Rates Decrease With Time. Shows the share of emails matched to a reply by day of week, aggregating across the four week-long snapshots.



B Additional Results

Table B.1: Changing Influence After the Onset of COVID-19.

	Scientist	Senior Leadership
Employee × March 12-20, 2020	0.99	1.04
	[0.90, 1.09]	[0.96, 1.12]
Employee × September 10-18, 2020	0.96	1.11
	[0.85, 1.08]	[1.02, 1.21]
Employee × March 11-19, 2021	0.94	1.03
	[0.83, 1.05]	[0.94, 1.12]
Employee	1.03	1.20
	[0.94, 1.14]	[1.12, 1.29]
March 12-20, 2020	1.21	1.16
	[1.17, 1.25]	[1.12, 1.21]
September 10-18, 2020	1.13	1.11
	[1.09, 1.17]	[1.06, 1.16]
March 11-19, 2021	1.09	1.08
	[1.05, 1.14]	[1.04, 1.13]
Sender is White	0.99	0.98
	[0.96, 1.02]	[0.95, 1.01]
Sender is Male	0.98	0.96
	[0.94, 1.01]	[0.93, 1.00]
Sender's Monthly Salary (1 SD)	1.08	1.05
	[1.06, 1.10]	[1.03, 1.07]
N	202 470	150 513
Hour-of-day Strata	✓	✓
Weekend Strata	✓	✓
Sender Years Exp. Strata	✓	✓
Receiver Strata	✓	✓